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Assignment 5

METCS544A2_S2024

Please type or handwrite your answers clearly and concisely. Show all work - credit will not be given for numerical solutions that appear without explanation. **[Total 120 = 3 + 117 pts]**

Grading Rubric

Each question is worth 3 points and will be graded as follows:

3 points: Correct answer with work shown

2 points: Incorrect answer but attempt shows some understanding (work shown)

1 point: Incorrect answer but an attempt was made (work shown)

0 points: Left blank or made little to no effort/work not shown

Reflective Journal [3 pts]

(link to your live google doc)

<https://docs.google.com/document/d/1dzAggy7UhyrmR9CylVDsXdToMj1JAtrVwGZaqPQtogo/edit?usp=sharing>

I. Collecting Data: Experimental Design (11 x 3 = 33 pts)

1) A local hospital compiles data on the length of time a patient is in surgery and the length of their stay in the hospital after surgery. A newbie hospital worker notices that the length of time a patient is in surgery is highly correlated with their hospital stay after. Explain to the newbie why this correlation does not imply a causation.

Answer:

The length of the surgery and the length of the hospital stay could be affected by a lurking variable such as the severity of the medical condition.

2) The good folks at Apples Inc want to create bags of pre-sliced apples for kids to easily put in their lunch boxes. They want to test different mixtures of preservatives (A, B, and C) on their Honeycrisp apples. They will treat all the apples in a bushel of Honeycrisps by randomly assigning them to each preservation treatment, and then comparing how long they are able to remain in the bag before they begin to brown.

(a) Identify the experimental units, the explanatory and response variables, and the treatments.

Answer:

Experimental units: Honeycrisp apples
Explanatory variable: preservative mixture
Response variable: time before browning
Treatment: preservation

(b) The researchers plan to use a completely randomized design. Describe how they should assign treatments to the 126 experimental units.

Answer:

For each experimental unit, they should roll a 6 sided die. If the die comes up 1 or 2, they assign it to treatment A, 3 or 4 to B, and 5 or 6 to C.

(c) The researchers suspect that the type of apple will influence how well the preservation method works. They want to repeat this experiment with 4 different types of apples. Describe how they should change the design of the experiment to account for this addition.

Answer:

They should make sure there are equal numbers of each type of apple with each type of preservative. They should block on apple type first then randomly assign to each preservative treatment.

3) A study was conducted to determine if taking a daily dose of aspirin reduces the chance of catching the common cold. This study was conducted using 550 volunteers who were not already on an aspirin regimen. The subjects were randomly assigned to one of two groups: a treatment group who received a low dose of aspirin daily, or a control group who received a placebo. The subjects were unaware of what group they were in. At the end of the study, the subjects were asked to meet with a doctor to discuss if they had any symptoms pertaining to the common cold.

(a) Is this study an experiment or an observational study? Explain your answer.

Answer:

This is an experiment because there is a treatment being applied. The researchers are influencing the outcome of the study by assigning some subjects to take aspirin and other to take placebo.

(b) What would be the advantage of having this study be double blind?

Answer:

If the study is double blind, the doctors who ask the subjects if they have cold-like symptoms won't be able to influence the results. If it's not double blind then the doctors might be more likely to indicate someone who is on placebo has cold symptoms even if they are the same symptoms as someone taking aspirin, or the other way around.

(c) Would blocking according to gender be worthwhile in this study? Explain your answer.

Answer:

To the best of my knowledge, aspirin does not affect men and women significantly differently. If each participant is given exactly the same dose then it would be worthwhile to separate the possibility of different subject weights affecting dosage. If the aspirin is dosed according to body weight (mg/kg) then there wouldn't be much advantage to blocking on gender.

(d) Describe what the "placebo effect" would look like in this experiment.

Answer:

Someone who thinks they are taking aspirin but is actually taking placebo would be less likely to report cold-like symptoms than if they knew they weren't taking aspirin.

4) Type 1 diabetes is thought to be caused by an autoimmune reaction (the body attacks itself by mistake) that destroys the cells in the pancreas that make insulin. It is a disease that currently has no cure. Researchers were to conduct a clinical trial on a promising new drug that helps the body produce its own insulin.

(a) You have 400 volunteers with Type 1 diabetes. Describe a completely randomized design for this experiment.

Answer:

Subjects are randomly assigned to either take the new drug or take a placebo.

(b) You have 250 children (17 and younger) with Type 1 diabetes (120 females and 130 males), as well as 300 adults (18 and older) with Type 1 diabetes (190 females and 110 males). Describe a completely randomized block design.

Answer:

Subjects are blocked by age first, then blocked by gender. This results in 4 blocks: female children, male children, female adults, male adults. The subjects in each block are randomly assigned to take the new drug or placebo.

(c) Explain why a matched pairs design would not work for this experiment.

Answer:

There are unequal number of subjects in each group. Not every subject would have a matched pair.

II. Collecting Data: Scope of Inference (5 x 3 = 15 pts)

Directions: For the study below, answer the questions that follow. Then, in the graphic organizer, make a check on the correct scope of the study.

Lack of sleep is associated with increased risk of cardiovascular disease, depression, and other health concerns. A new study now shows that lack of sleep also affects social interactions, making people less willing to help others (empathy). Scientists placed 24 healthy volunteers in a functional magnetic resonance imager (fMRI) to scan their brains after eight hours of sleep and then again after a night of no sleep. The order that they received the treatment of sleep vs no sleep was randomly selected with the flip of a coin. They found that areas of the brain that form the theory of mind network, which is engaged when people empathize with others or try to understand other people's wants and needs, were less active after a sleepless night. "Here, we found that a decrease in the quality of someone's sleep from one night to the next predicted a significant decrease in the desire to help other people from one subsequent day to the next."

Source: <https://www.sciencedaily.com/releases/2022/08/220823143827.htm>

Is this an observational study or an experiment? Explain. This is an experiment because a treatment is being imposed on the test subjects, either sleep or no sleep.	Which box of inference does this study fall under? <table border="1" data-bbox="987 667 1494 1165"> <tr> <td></td> <td>Inference about cause and effect</td> <td>No Inference about cause and effect</td> </tr> <tr> <td>Inference about the Population</td> <td></td> <td></td> </tr> <tr> <td>No Inference about the Population</td> <td>✓</td> <td></td> </tr> </table>		Inference about cause and effect	No Inference about cause and effect	Inference about the Population			No Inference about the Population	✓	
		Inference about cause and effect	No Inference about cause and effect							
Inference about the Population										
No Inference about the Population		✓								
Does this study involve random sampling or random assignment? Explain. This study involves random assignment because the subjects were randomly assigned which night of sleep they got first, full eight hours or none at all.										
Can this study make inferences about the population or only the study participants? Explain. The study can only make inferences about the study participants because they are volunteers so they might not represent the whole population.										
Can this study make inferences about cause and effect or just an association between variables? Explain. This study can make inferences about cause and effect as long as it is well controlled to separate the explanatory variable from all other confounding or lurking variables.										

III. Probability Basics (12 x 3 = 36 pts)

Write all probabilities as decimals and round to the nearest three decimal places when needed.

1) You pick 1 marble from a bag that contains 8 marbles (2 blue, 3 red, and 3 yellow). Find the following probabilities.

a) P(Blue)

0.25

b) P(Yellow)

0.375

c) P(Green)

0

d) P(Blue or Yellow)

0.625

e) P(not Blue)

0.75

f) P(Red, Blue, or Yellow)

1

2) Below are the probabilities of pulling out a particular color from an M&M bag.

- a) What is the probability of pulling a blue M&M?
- b) Describe the likelihood of this event.

Brown	Red	Yellow	Green	Orange	Blue
0.30	0.20	0.20	0.10	0.10	?

Answer: a)

0.1

b)

The event is unlikely.

3) A survey of 324 people asked what their favorite food was. The results are shown below.

	Pizza	Burgers	Fried Chicken	Other	Total
Less than 18	60	23	5	34	122
18 and older	45	33	20	104	202
Total	105	56	25	138	324

If we randomly select a person from this sample,

a) What is the probability that person liked fried chicken?

0.077

b) What is the probability that person is less than 18 years old and likes burgers?

0.102

c) What is the probability that person likes pizza or burgers?

0.497

d) What is the probability that person is less than 18 or 18 and older?

1

IV. The Addition Rule (6 x 3 = 18 pts)

A standard deck of cards has 52 cards. Each deck has 4 different suits: Clubs (black), Spades (black), Diamonds (red), and Hearts (red). Each suit contains the following cards: Ace, 2, 3, 4, 5, 6, 7, 8, 9, 10, Jack, Queen, and a King.

1) Consider the events listed below and a fair deck of cards.

C = draw a heart

D = draw a face card

E = draw an ace

Which of the following are mutually exclusive? Explain.

a) C and D

b) C and E

c) D and E

Not mutually exclusive, can draw the Queen of Hearts.

Not mutually exclusive, can draw the Ace of Hearts.

Mutually exclusive, a card cannot be both an ace and a face card at the same time.

2) One six-sided die is rolled. What is the probability that the die will be ...

a) Factor of 12 *or* Factor of 9

5 is the only die face not a factor of 12 or 9 so the probability is $1/6$ or 0.167.

b) Less than 3 *or* Greater than 5

4 is the only die face not less than 3 or greater than 5 so the probability is $1/6$ or 0.167.

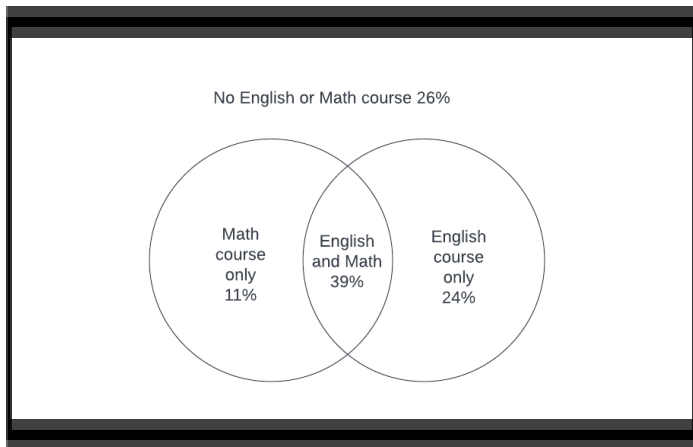
3) The probability that a student owns a car is 0.65, and the probability that a student owns a computer is 0.82. If the probability that a student owns both is 0.55, what is the probability that a randomly selected student owns a car or computer?

$$0.65 + 0.82 - 0.55 = 0.92$$

V. Venn Diagrams, Unions, and Intersections (5 x 3 = 15 pts)

1) At a liberal arts college in the Midwest, 39% of freshmen are enrolled in both a math course and an English course, 11% are enrolled in a math course, but not an English course, and 63% are enrolled in an English course.

a) Construct a Venn diagram to illustrate this situation. Be sure to label all sets.



b) What is the probability that a freshman student selected at random is taking an English course but not a mathematics course?

24%

c) What is the probability a student is not enrolled in either course?

26%

2) Create a table to display all data that is described.

M = Male

F = Female

D = owns a desktop

L = owns a laptop

B = owns both

N = owns neither

$$P(F \cap D) = \frac{21}{235}$$

$$P(M) = \frac{153}{235}$$

$$P(L) = \frac{106}{235}$$

$$P(M \cap L) = \frac{52}{235}$$

$$P(F \cap N) = \frac{6}{235}$$

$$P(N) = \frac{10}{235}$$

$$P(F \cup D) = \frac{86}{235}$$

	Desktop	Laptop	Both	Neither	Total
Female	21	54	1	6	82
Male	4	52	93	4	153
Total	25	106	94	10	235

3) The probability of rain on Saturday or Sunday is given as 0.5. The probability of rain on Saturday is 0.2 and the probability of rain on Sunday is 0.4. What is the probability that it will rain on both Saturday and Sunday?

$$0.2 + 0.4 - 0.5 = 0.1$$

THE END